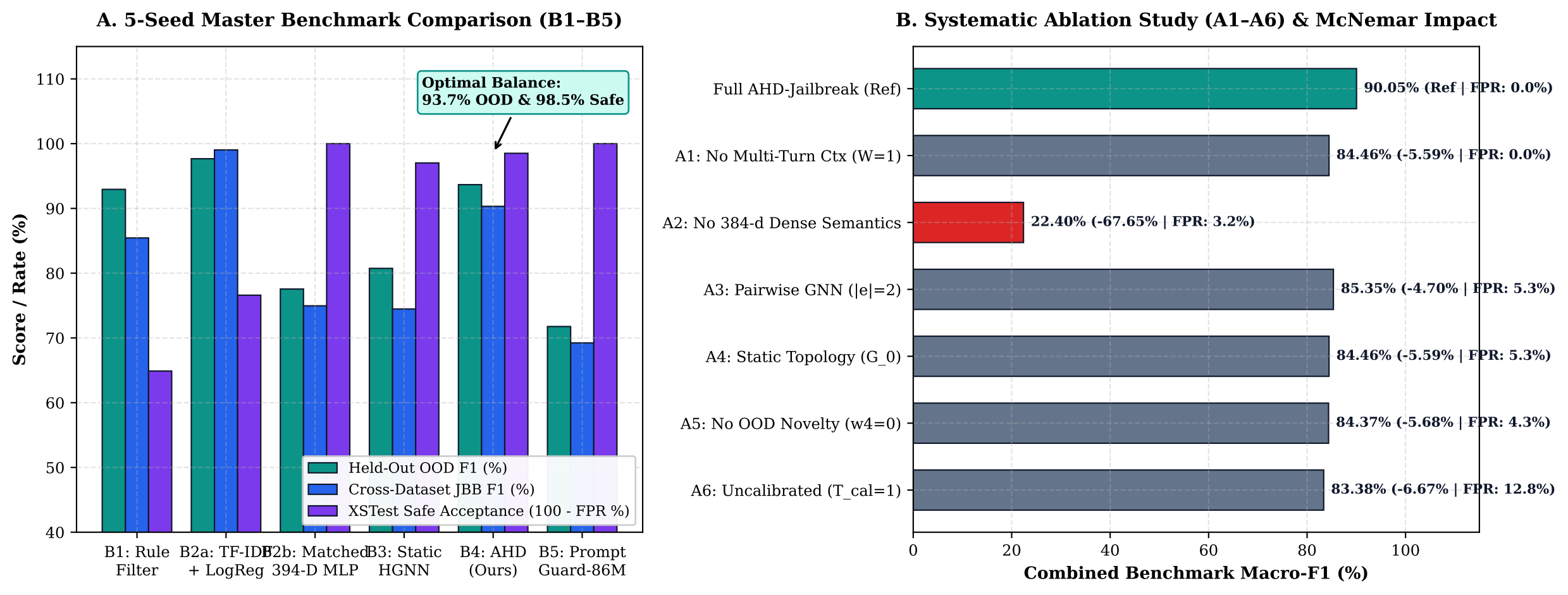
A. 5-Seed Master Benchmark Comparison (B1–B5)
Score / Rate (%)
Optimal Balance:
93.7% OOD & 98.5% Safe
Held-Out OOD F1 (%)
Cross-Dataset JBB F1 (%)
XSTest Safe Acceptance (100 - FPR %)
B1: Rule Filter
B2a: TF-IDF + LogReg
B2b: Matched 394-D MLP
B3: Static HGNN
B4: AHD (Ours)
B5: Prompt Guard-86M
B. Systematic Ablation Study (A1–A6) & McNemar Impact
Full AHD-Jailbreak (Ref)
90.05% (Ref | FPR: 0.0%)
A1: No Multi-Turn Ctx (W=1)
84.46% (-5.59% | FPR: 0.0%)
A2: No 384-d Dense Semantics
22.40% (-67.65% | FPR: 3.2%)
A3: Pairwise GNN (|e|=2)
85.35% (-4.70% | FPR: 5.3%)
A4: Static Topology (G_0)
84.46% (-5.59% | FPR: 5.3%)
A5: No OOD Novelty (w4=0)
84.37% (-5.68% | FPR: 4.3%)
A6: Uncalibrated (T_cal=1)
83.38% (-6.67% | FPR: 12.8%)
Combined Benchmark Macro-F1 (%)